

Technical Project Report: Dynamic Speed Limiter for Automobiles

RF-Based Localized Electronic Throttle Interception System

1. Executive Summary & Concept Validation

The **Dynamic Speed Limiter for Automobiles** is an infrastructure-to-vehicle (I2V) wireless speed control system. It divides transit corridors into **Free Zones** (unrestricted driving) and **Speed Limit Zones** (school zones, municipal limits, hazardous curves) using localized Sub-GHz RF beacons.

When a vehicle enters a restricted zone, an on-board interception unit receives the broadcasted speed limit packet, monitors the vehicle's speed via an optocoupler-isolated Vehicle Speed Sensor (VSS) signal, and actively caps the electronic throttle signal sent to the Engine Control Unit (ECU) if the driver exceeds the safe threshold.

2. System Architecture & Component Selection

The architecture consists of two main hardware units:

- 1. **Roadside Transmitter Node (Infrastructure):** Continuously broadcasts Sub-GHz (433 MHz) encoded zone parameters containing maximum permissible velocity data.
- 2. **On-Board Vehicle Interception Node:** Receives RF packets, reads the driver's dual analog pedal potentiometers, computes velocity feedback, and drives dual 12-bit DACs to supply safe throttle levels to the vehicle ECU.

Hardware Selection Matrix

Sub-System Node	Component Selected	Primary Function & Justification
Microcontroller	Robozar ESP32 Dev Board	Dual-core 240MHz architecture isolates real-time VSS interrupt loops from LoRa RF packet processing.
RF Transceiver	xcluma SX1278 LoRa (433MHz)	Sub-GHz spread-spectrum modulation provides strong penetration through foliage, bad weather, and vehicle bodywork.
Throttle Interface	Dual MCP4725 12-Bit DACs	Generates true analog voltage outputs (0--5V) across I2C addresses 0x60 and 0x61

Sub-System Node	Component Selected	Primary Function & Justification
		to emulate dual pedal tracks without PWM noise.
Output Stage	LM358 Automotive Op-Amp	Unity-gain buffer configuration provides impedance matching and prevents ECU input impedance loading.
Speed Sensor Feed	PC817 Optocoupler	5kV galvanic isolation decouples noisy 12V automotive VSS pulses from 3.3V ESP32 GPIO pins.
Power Stage	MP1584EN Buck Converter	Operates at > 90% thermal efficiency to step down noisy 12V--14.4V battery power to a stable 5V system rail.

3. Control Theory & Mathematical Modeling

The interception algorithm modifies driver input without abruptly shutting off engine power:

$$\text{Speed Error: } E(t) = V_{\text{current}} - V_{\text{limit}}$$

$$\text{Output Modifier: } M = 1.0 - (E(t) \times K_p)$$

$$\text{Final ECU Voltage: } V_{\text{ECU}} = \max(V_{\text{floor}}, V_{\text{driver}} \times M)$$

- V_{current} : Live vehicle velocity calculated via VSS interrupt pulse frequency.
- V_{limit} : Active speed limit received from the LoRa roadside beacon.
- K_p : Proportional damping factor (0.04) tuned to avoid abrupt engine braking.
- V_{floor} : Safety minimum throttle floor (15%) to ensure power steering and power brakes remain fully active even under maximum speed clamping.

4. Hardware Wiring & Interconnect Layout

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[ 14.4V Battery ] ---> [ MP1584EN Buck ] ---> 5V Rail ---> ESP32 / MCP4725 / LM358
                        |
[ 12V VSS Pulse ] ---> [ PC817 Opto ] -----> GPIO 4 (ESP32 Interrupt)

[ LoRa Antenna ] ---> [ SX1278 Module ] ---> SPI Pins (GPIO 5, 18, 19, 23)

[ Pedal Track A ] -----> GPIO 34 (Analog Read A)
[ Pedal Track B ] -----> GPIO 35 (Analog Read B)

ESP32 (I2C) -----> [ Dual MCP4725 DACs ] ---> [ LM358 Op-Amp ] ---> Vehicle ECU

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Firmware Control Flow

1. **System Initialization:** Configures I2C bus (400 kHz), attaches hardware interrupt on GPIO 4 for VSS pulses, and sets up LoRa radio frequency (433 MHz).
2. **Pulse Frequency Calculation:** Measures time elapsed between hardware interrupts to compute vehicle speed in km/h.
3. **RF Packet Verification:** Listens for incoming LoRa packets. If a valid zone limit packet is received, updates internal limit parameters.
4. **Throttle Modification:** Computes the modifier ratio M , adjusts pedal analog values, and writes output values to the dual MCP4725 DACs.
5. **Fail-Safe Operation:** If LoRa communication drops or invalid packets arrive, the system safely defaults to Free Zone mode, smoothly restoring full driver control.